# Supplementary material for UAI 2026 Submission 642
# LiRA vs. BaVarIA: per-testbed tables and ROC curves (32-replicate mean)

LiRA vs. BaVarIA-$n$ vs. BaVarIA-$t$ comparison across 12 testbeds, seven shadow-model budgets ($K \in \{4, 8, 16, 32, 64, 128, 254\}$), and 32 leave-one-out replicates per configuration (every shadow model audited as the held-out target). Antithetic-paired design (128 pairs, $K = 254$ max). Online uses both IN and OUT shadow observations; offline replaces the IN class with a global prior. ROC curves are interpolated onto a common log-spaced FPR grid and aggregated with the mean; bands show $\pm 1$ SE.

## 1 Online results

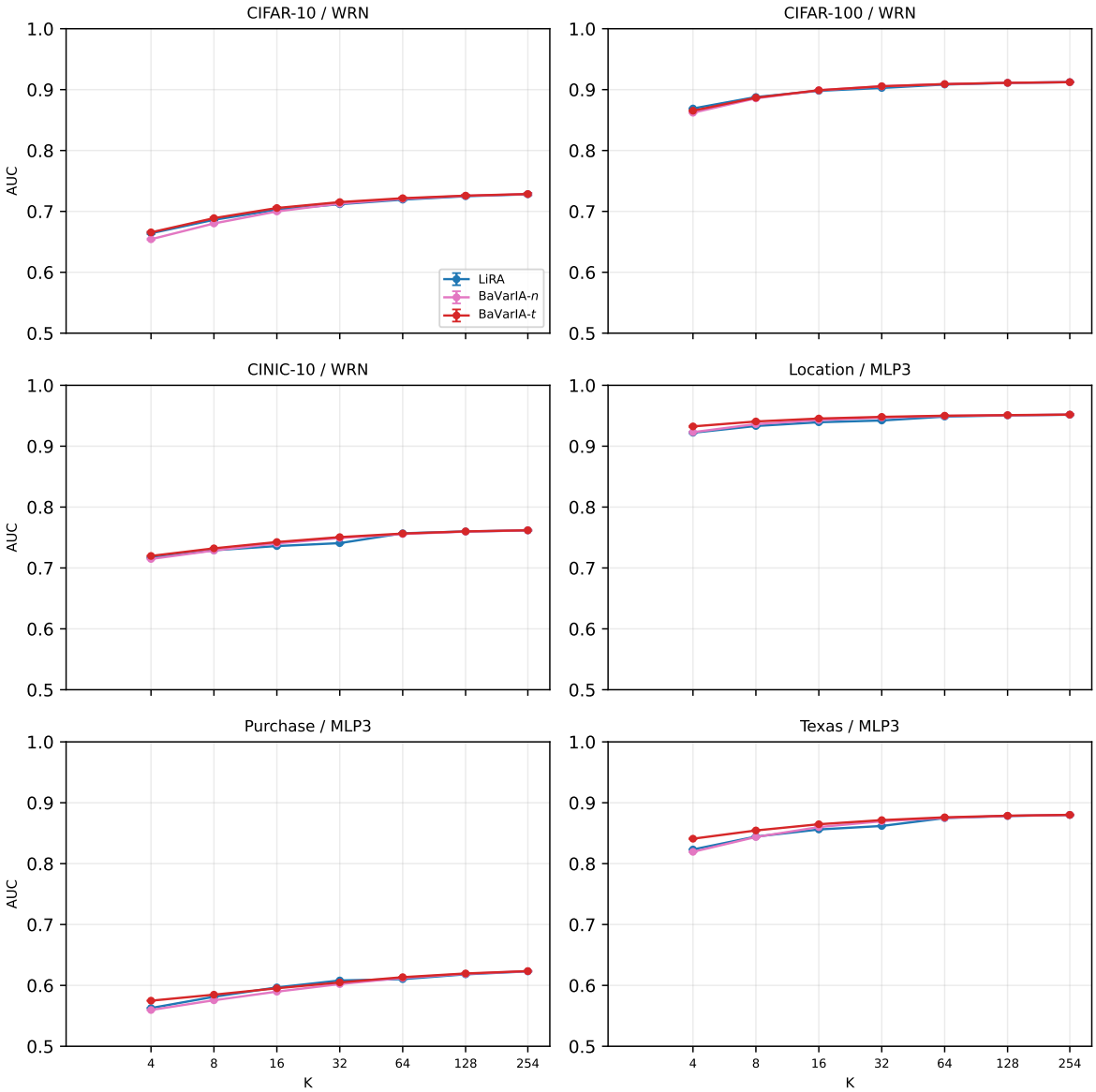


Figure 1: Online AUC vs. $K$ (WRN / MLP3 testbeds). Mean $\pm 1$ SE over 32 replicates.

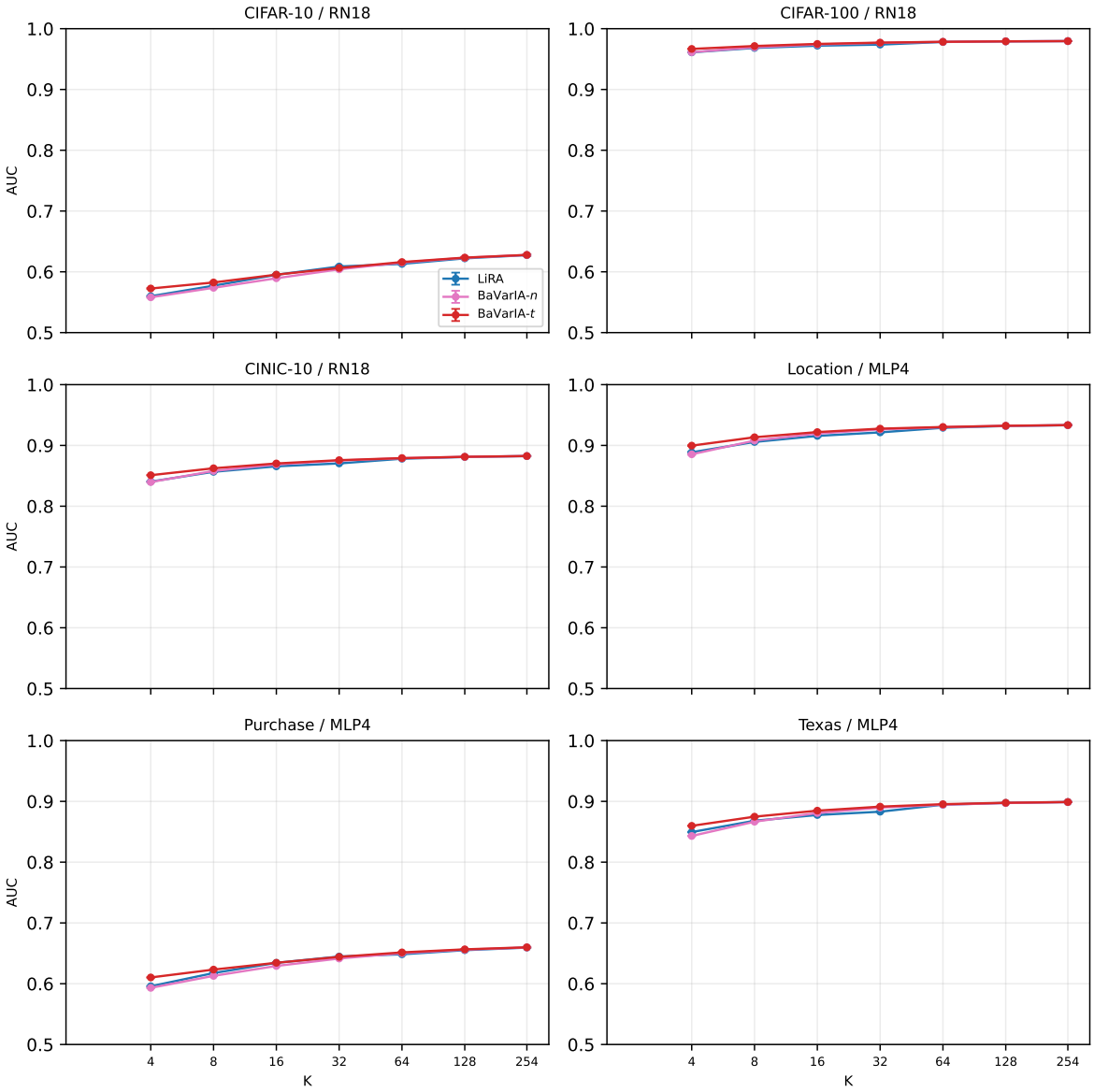


Figure 2: Online AUC vs. $K$ (RN18 / MLP4 testbeds). Mean $\pm 1$ SE over 32 replicates.

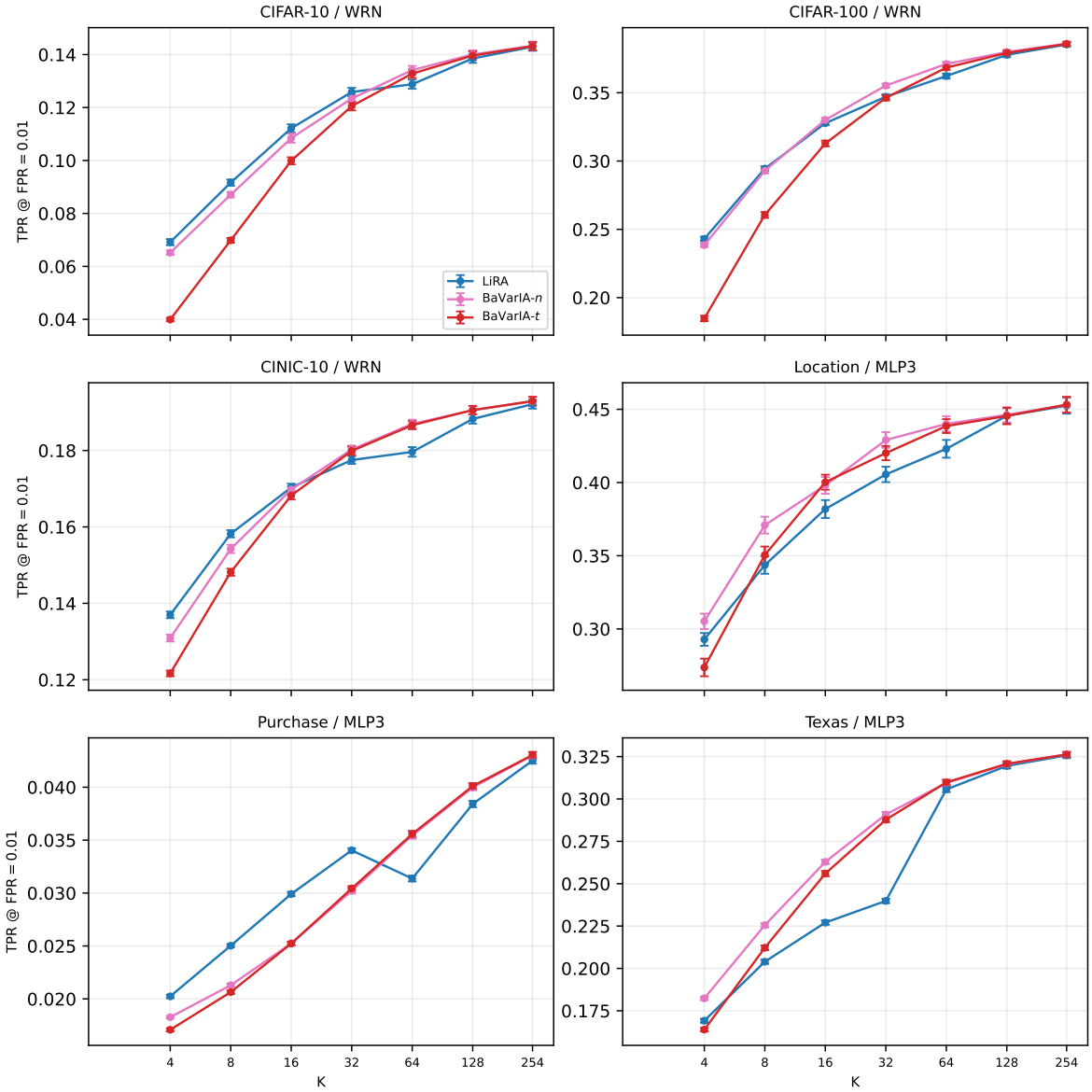


Figure 3: Online TPR at FPR= 0.01 vs. $K$ (WRN / MLP3 testbeds). Mean $\pm 1$ SE over 32 replicates.

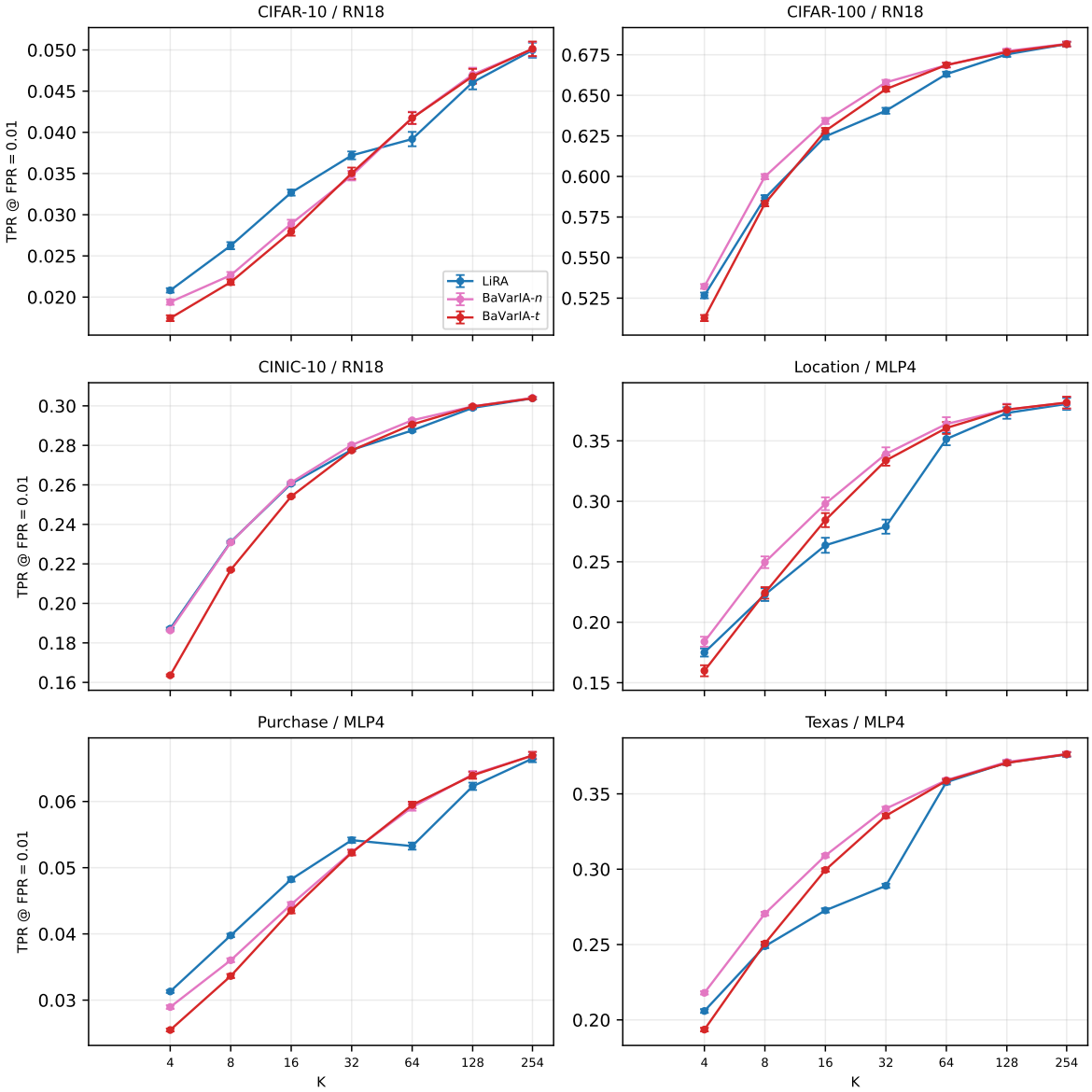


Figure 4: Online TPR at FPR= 0.01 vs. $K$ (RN18 / MLP4 testbeds). Mean $\pm 1$ SE over 32 replicates.

**Online results – ROC curves**

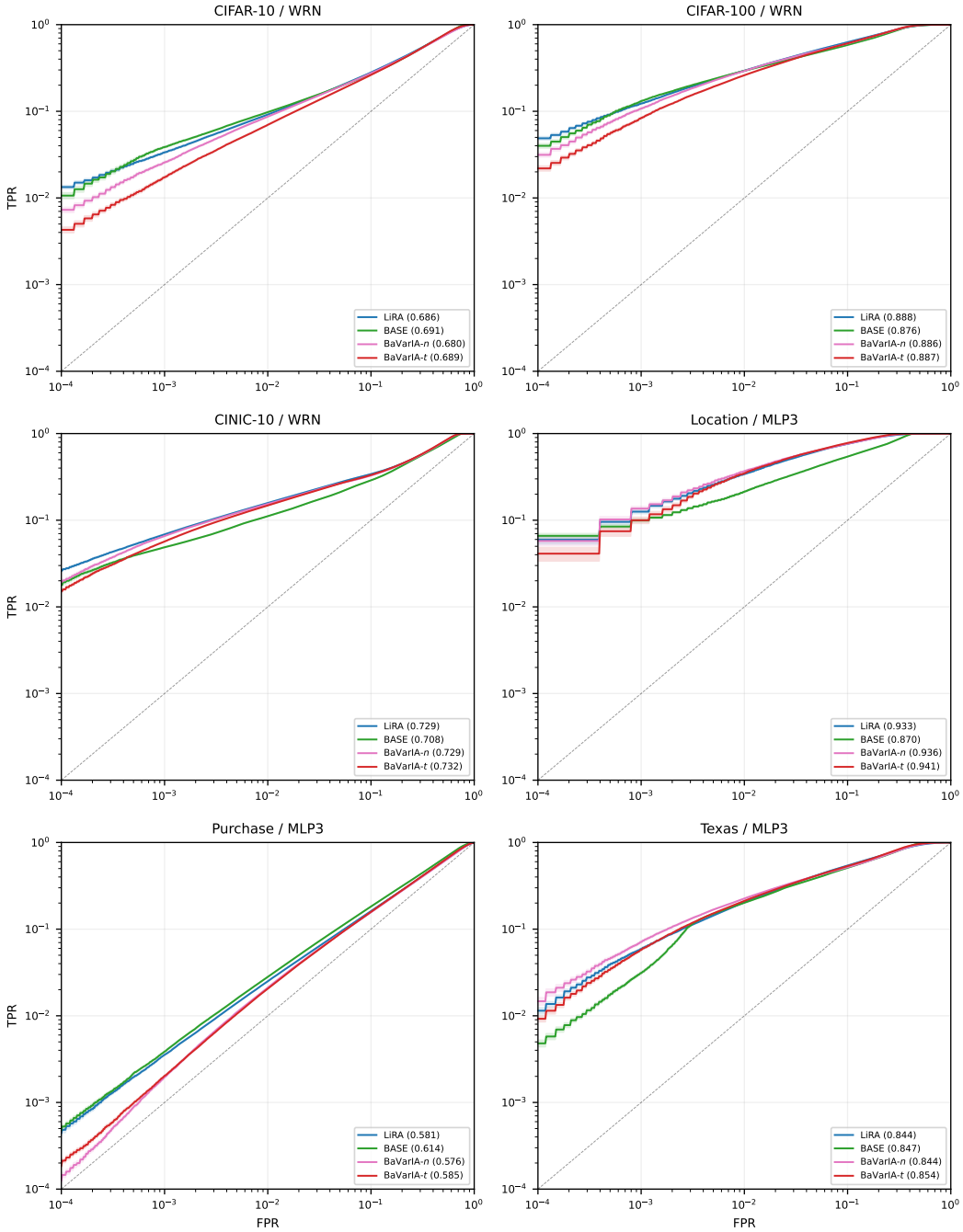


Figure 5: Online ROC at $K = 8$ (WRN / MLP3 testbeds). LiRA, BASE, BaVarIA-$n$, BaVarIA-$t$ overlaid.

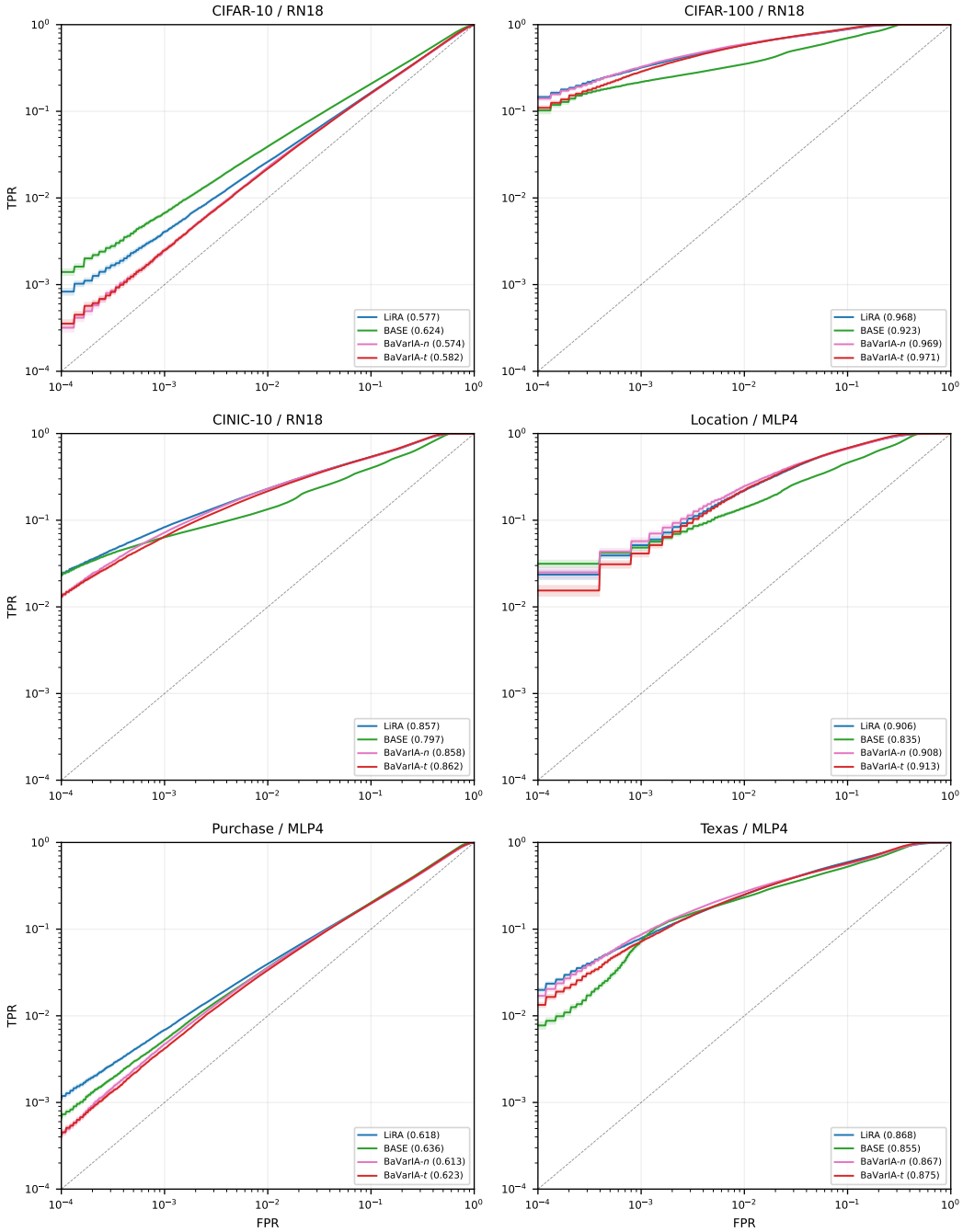


Figure 6: Online ROC at $K = 8$ (RN18 / MLP4 testbeds). LiRA, BASE, BaVarIA-$n$, BaVarIA-$t$ overlaid.

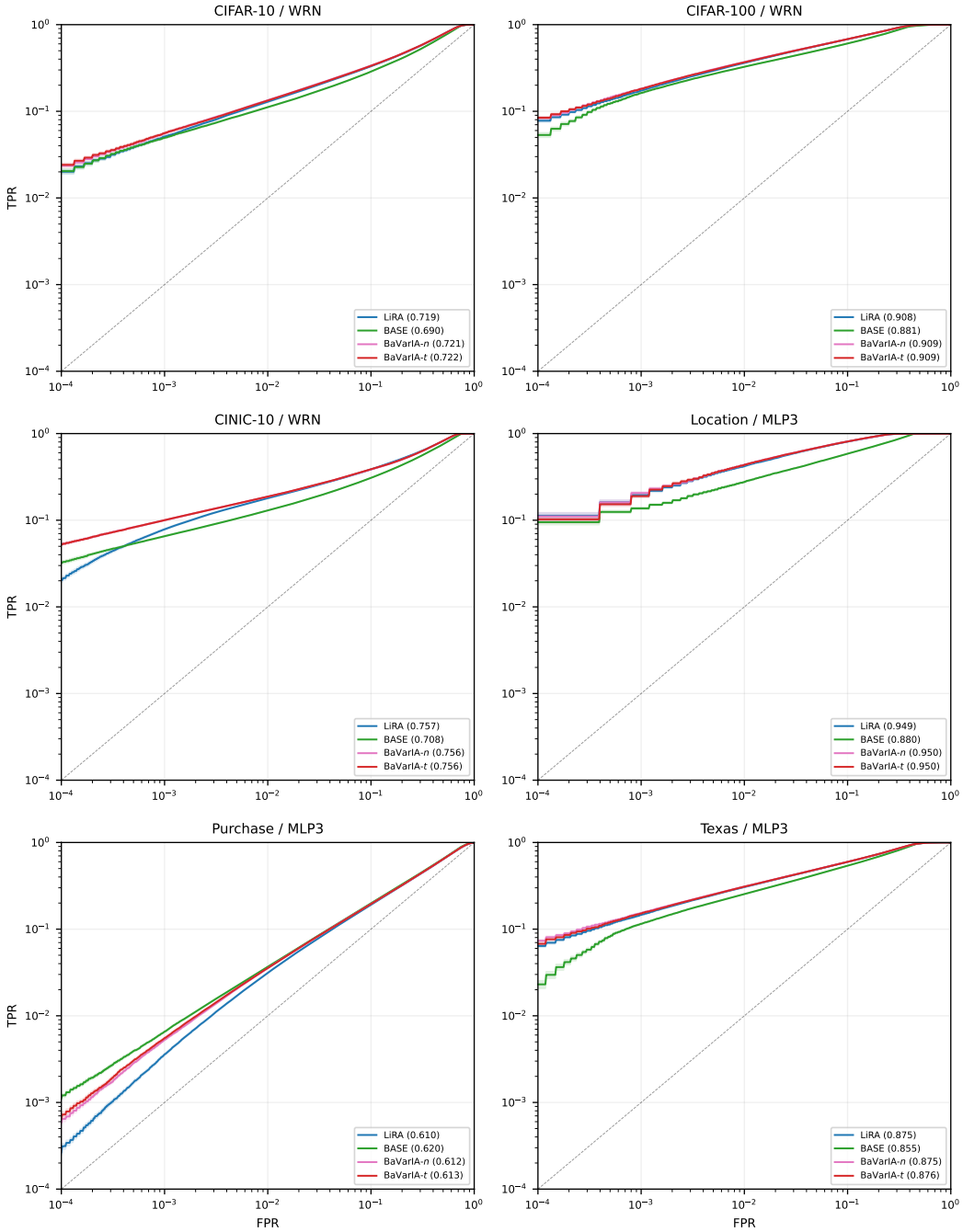


Figure 7: Online ROC at $K = 64$ (WRN / MLP3 testbeds). LiRA, BASE, BaVarIA-$n$, BaVarIA-$t$ overlaid.

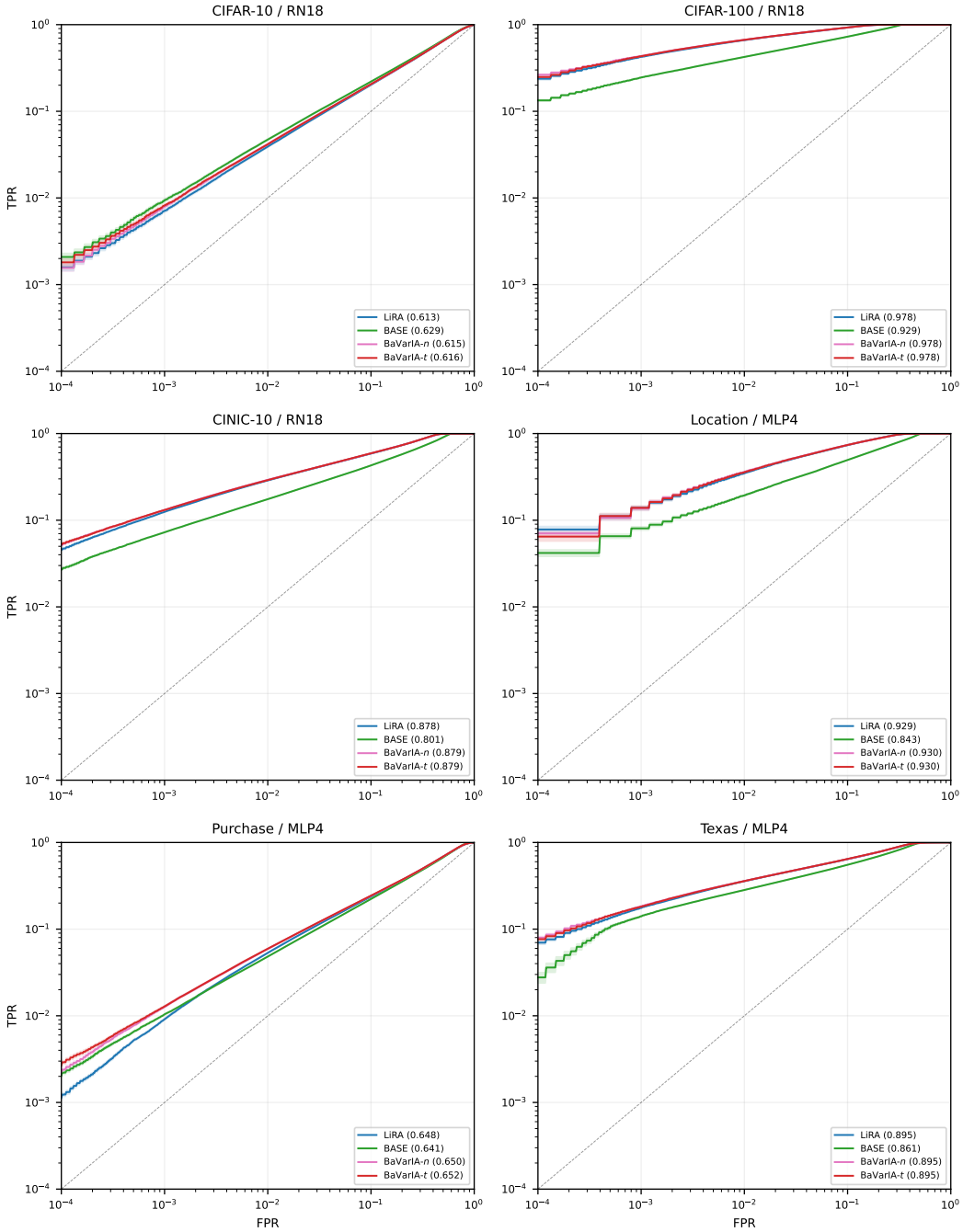


Figure 8: Online ROC at $K = 64$ (RN18 / MLP4 testbeds). LiRA, BASE, BaVarIA-$n$, BaVarIA-$t$ overlaid.

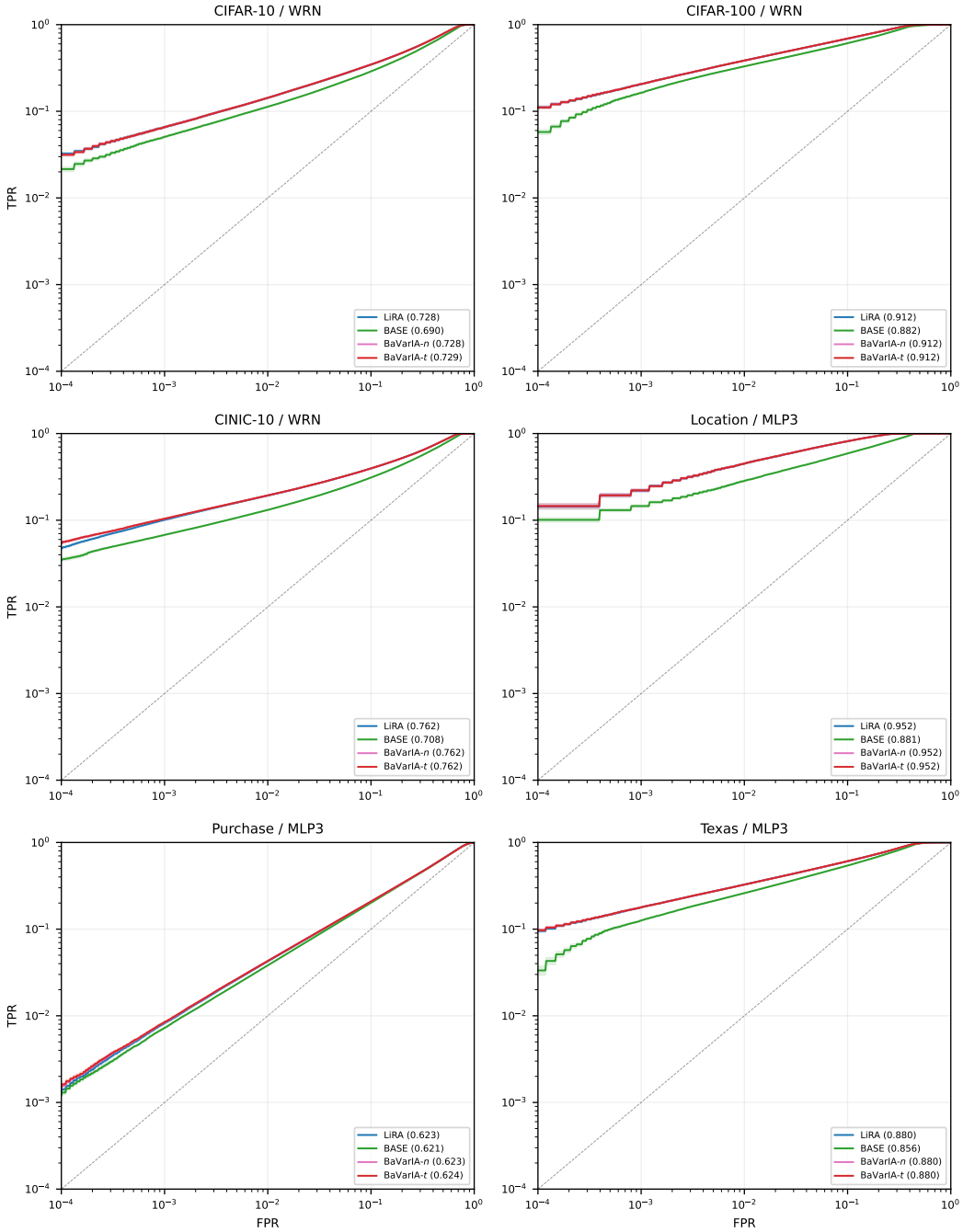


Figure 9: Online ROC at $K = 254$ (WRN / MLP3 testbeds). LiRA, BASE, BaVarIA-$n$, BaVarIA-$t$ overlaid.

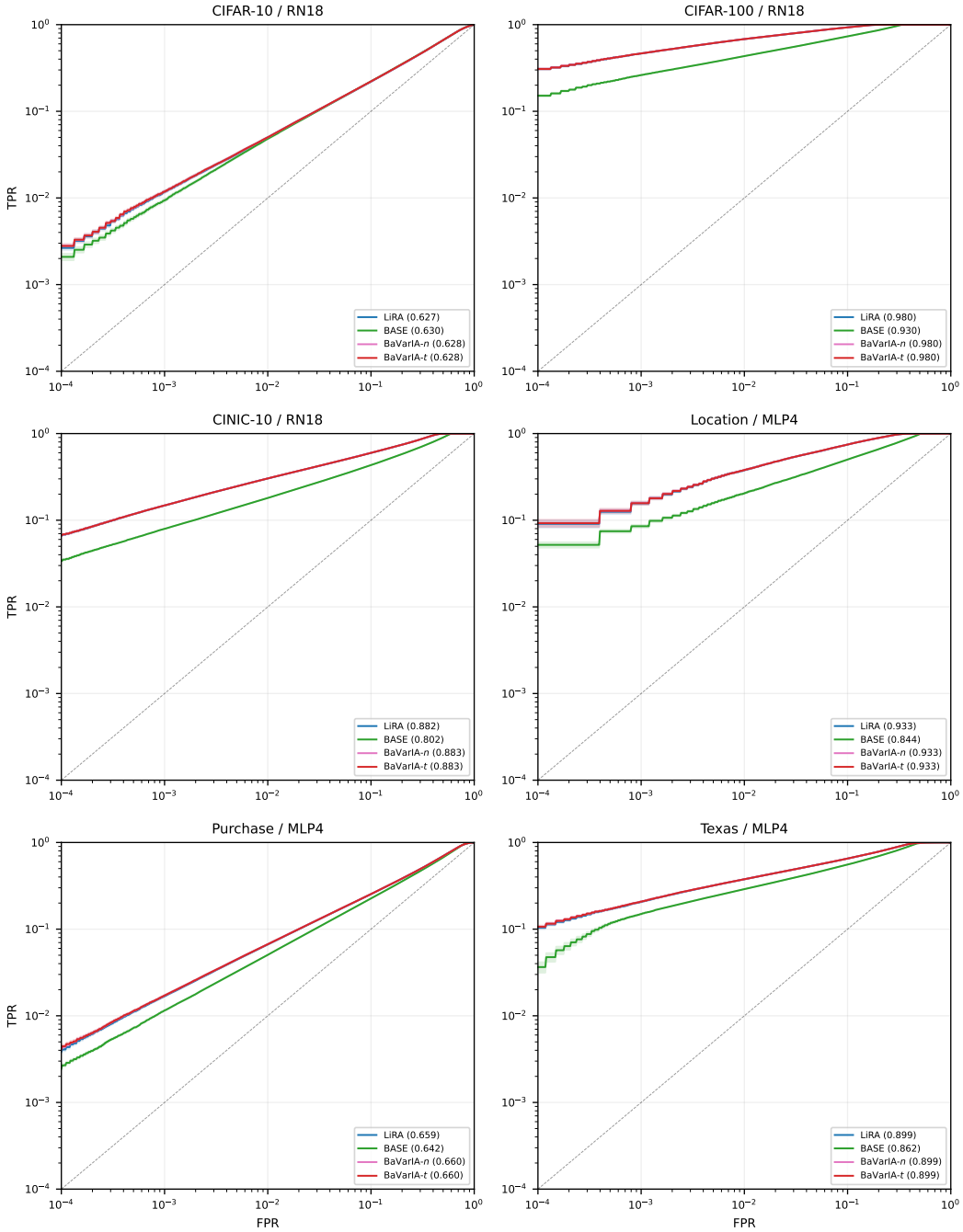


Figure 10: Online ROC at $K = 254$ (RN18 / MLP4 testbeds). LiRA, BASE, BaVarIA-$n$, BaVarIA-$t$ overlaid.

### Online results – per-testbed tables

| Testbed | LiRA | BaVarIA-$n$ | BaVarIA-$t$ |
|---|---|---|---|
| CIFAR-10 / WRN | 0.686 | 0.680 | **0.689** |
| CIFAR-100 / WRN | **0.888** | 0.886 | 0.887 |
| CINIC-10 / WRN | 0.729 | 0.729 | **0.732** |
| Location / MLP3 | 0.933 | 0.936 | **0.941** |
| Purchase / MLP3 | 0.581 | 0.576 | **0.585** |
| Texas / MLP3 | 0.844 | 0.844 | **0.854** |
| CIFAR-10 / RN18 | 0.577 | 0.574 | **0.582** |
| CIFAR-100 / RN18 | 0.968 | 0.969 | **0.971** |
| CINIC-10 / RN18 | 0.857 | 0.858 | **0.862** |
| Location / MLP4 | 0.906 | 0.908 | **0.913** |
| Purchase / MLP4 | 0.618 | 0.613 | **0.623** |
| Texas / MLP4 | 0.868 | 0.867 | **0.875** |

Table 1: Online AUC, $K = 8$. Mean over 32 replicates. Best per row in bold.

| Testbed | LiRA | BaVarIA-$n$ | BaVarIA-$t$ |
|---|---|---|---|
| CIFAR-10 / WRN | 0.719 | 0.721 | **0.722** |
| CIFAR-100 / WRN | 0.908 | **0.909** | **0.909** |
| CINIC-10 / WRN | **0.757** | 0.756 | 0.756 |
| Location / MLP3 | 0.949 | 0.950 | **0.950** |
| Purchase / MLP3 | 0.610 | 0.612 | **0.613** |
| Texas / MLP3 | 0.875 | 0.875 | **0.876** |
| CIFAR-10 / RN18 | 0.613 | 0.615 | **0.616** |
| CIFAR-100 / RN18 | 0.978 | 0.978 | **0.978** |
| CINIC-10 / RN18 | 0.878 | 0.879 | **0.879** |
| Location / MLP4 | 0.929 | 0.930 | **0.930** |
| Purchase / MLP4 | 0.648 | 0.650 | **0.652** |
| Texas / MLP4 | 0.895 | 0.895 | **0.895** |

Table 2: Online AUC, $K = 64$. Mean over 32 replicates. Best per row in bold.

| Testbed | LiRA | BaVarIA-$n$ | BaVarIA-$t$ |
|---|---|---|---|
| CIFAR-10 / WRN | 0.728 | 0.728 | **0.729** |
| CIFAR-100 / WRN | **0.912** | **0.912** | **0.912** |
| CINIC-10 / WRN | 0.762 | 0.762 | **0.762** |
| Location / MLP3 | 0.952 | 0.952 | **0.952** |
| Purchase / MLP3 | 0.623 | 0.623 | **0.624** |
| Texas / MLP3 | 0.880 | 0.880 | **0.880** |
| CIFAR-10 / RN18 | 0.627 | 0.628 | **0.628** |
| CIFAR-100 / RN18 | **0.980** | **0.980** | **0.980** |
| CINIC-10 / RN18 | **0.882** | **0.883** | **0.883** |
| Location / MLP4 | 0.933 | 0.933 | **0.933** |
| Purchase / MLP4 | 0.659 | 0.660 | **0.660** |
| Texas / MLP4 | **0.899** | 0.899 | **0.899** |

Table 3: Online AUC, $K = 254$. Mean over 32 replicates. Best per row in bold.

| Testbed | LiRA | BaVarIA-$n$ | BaVarIA-$t$ |
|---|---|---|---|
| CIFAR-10 / WRN | **0.092** | 0.087 | 0.070 |
| CIFAR-100 / WRN | **0.294** | 0.293 | 0.261 |
| CINIC-10 / WRN | **0.158** | 0.154 | 0.148 |
| Location / MLP3 | 0.344 | **0.371** | 0.351 |
| Purchase / MLP3 | **0.025** | 0.021 | 0.021 |
| Texas / MLP3 | 0.204 | **0.226** | 0.212 |
| CIFAR-10 / RN18 | **0.026** | 0.023 | 0.022 |
| CIFAR-100 / RN18 | 0.587 | **0.600** | 0.583 |
| CINIC-10 / RN18 | **0.231** | 0.231 | 0.217 |
| Location / MLP4 | 0.223 | **0.250** | 0.224 |
| Purchase / MLP4 | **0.040** | 0.036 | 0.034 |
| Texas / MLP4 | 0.249 | **0.271** | 0.251 |

Table 4: Online TPR @ FPR= 0.01, $K = 8$. Mean over 32 replicates. Best per row in bold.

| Testbed | LiRA | BaVarIA-$n$ | BaVarIA-$t$ |
|---|---|---|---|
| CIFAR-10 / WRN | 0.129 | **0.134** | 0.133 |
| CIFAR-100 / WRN | 0.362 | **0.371** | 0.368 |
| CINIC-10 / WRN | 0.180 | **0.187** | 0.187 |
| Location / MLP3 | 0.423 | **0.440** | 0.439 |
| Purchase / MLP3 | 0.031 | 0.035 | **0.036** |
| Texas / MLP3 | 0.306 | 0.309 | **0.310** |
| CIFAR-10 / RN18 | 0.039 | **0.042** | **0.042** |
| CIFAR-100 / RN18 | 0.663 | 0.669 | **0.669** |
| CINIC-10 / RN18 | 0.288 | **0.293** | 0.291 |
| Location / MLP4 | 0.352 | **0.364** | 0.361 |
| Purchase / MLP4 | 0.053 | 0.059 | **0.059** |
| Texas / MLP4 | 0.358 | **0.359** | 0.359 |

Table 5: Online TPR @ FPR= 0.01, $K = 64$. Mean over 32 replicates. Best per row in bold.

| Testbed | LiRA | BaVarIA-$n$ | BaVarIA-$t$ |
|---|---|---|---|
| CIFAR-10 / WRN | 0.143 | **0.143** | 0.143 |
| CIFAR-100 / WRN | 0.385 | **0.386** | 0.386 |
| CINIC-10 / WRN | 0.192 | **0.193** | **0.193** |
| Location / MLP3 | 0.452 | 0.453 | **0.453** |
| Purchase / MLP3 | 0.043 | **0.043** | **0.043** |
| Texas / MLP3 | 0.326 | **0.326** | **0.326** |
| CIFAR-10 / RN18 | 0.050 | **0.050** | **0.050** |
| CIFAR-100 / RN18 | 0.682 | **0.682** | 0.682 |
| CINIC-10 / RN18 | 0.304 | **0.304** | 0.304 |
| Location / MLP4 | 0.381 | 0.381 | **0.382** |
| Purchase / MLP4 | 0.066 | **0.067** | **0.067** |
| Texas / MLP4 | 0.376 | **0.376** | 0.376 |

Table 6: Online TPR @ FPR= 0.01, $K = 254$. Mean over 32 replicates. Best per row in bold.

## 2 Offline results

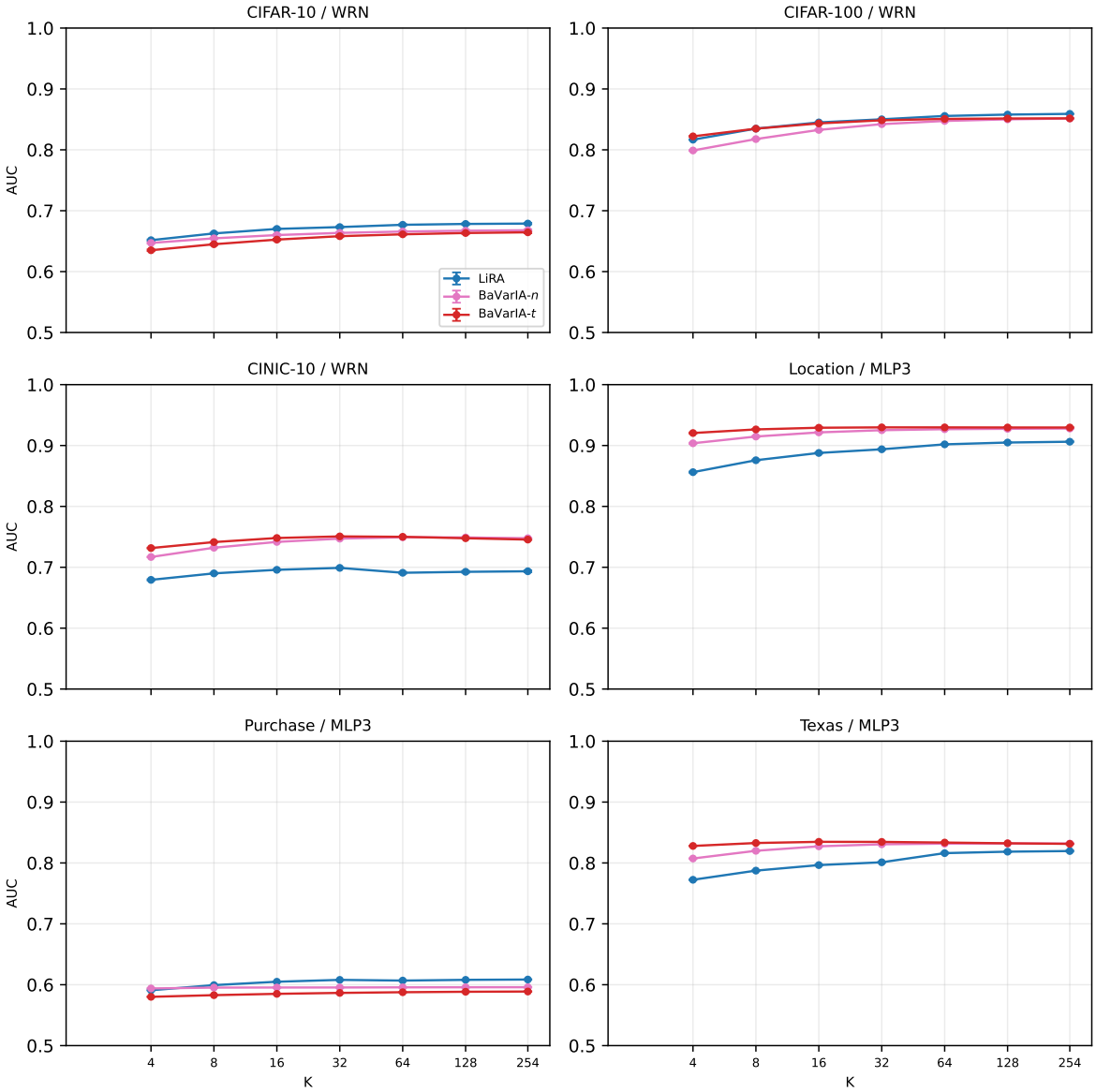


Figure 11: Offline AUC vs. $K$ (WRN / MLP3 testbeds). Mean $\pm 1$ SE over 32 replicates.

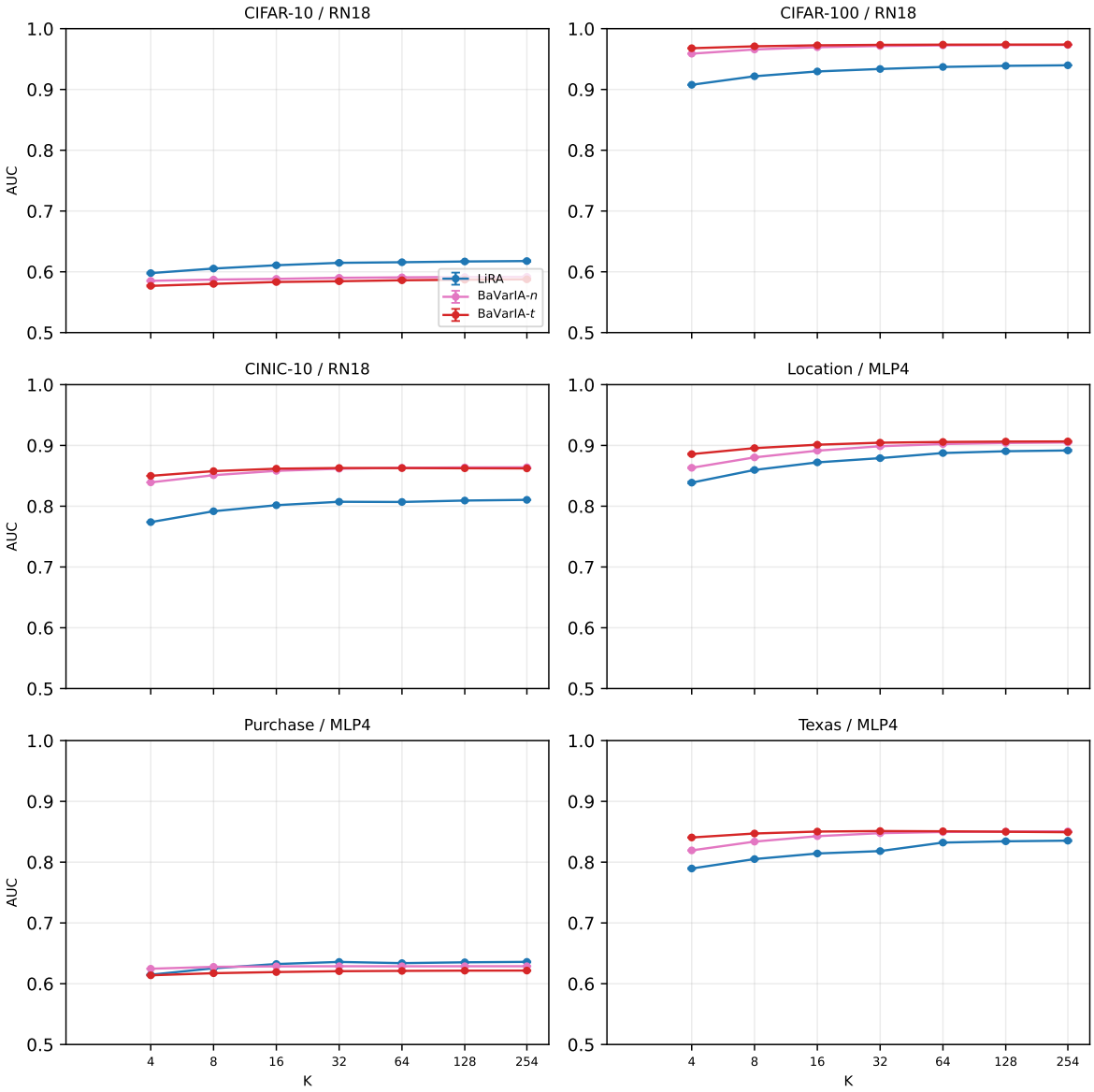


Figure 12: Offline AUC vs. $K$ (RN18 / MLP4 testbeds). Mean $\pm 1$ SE over 32 replicates.

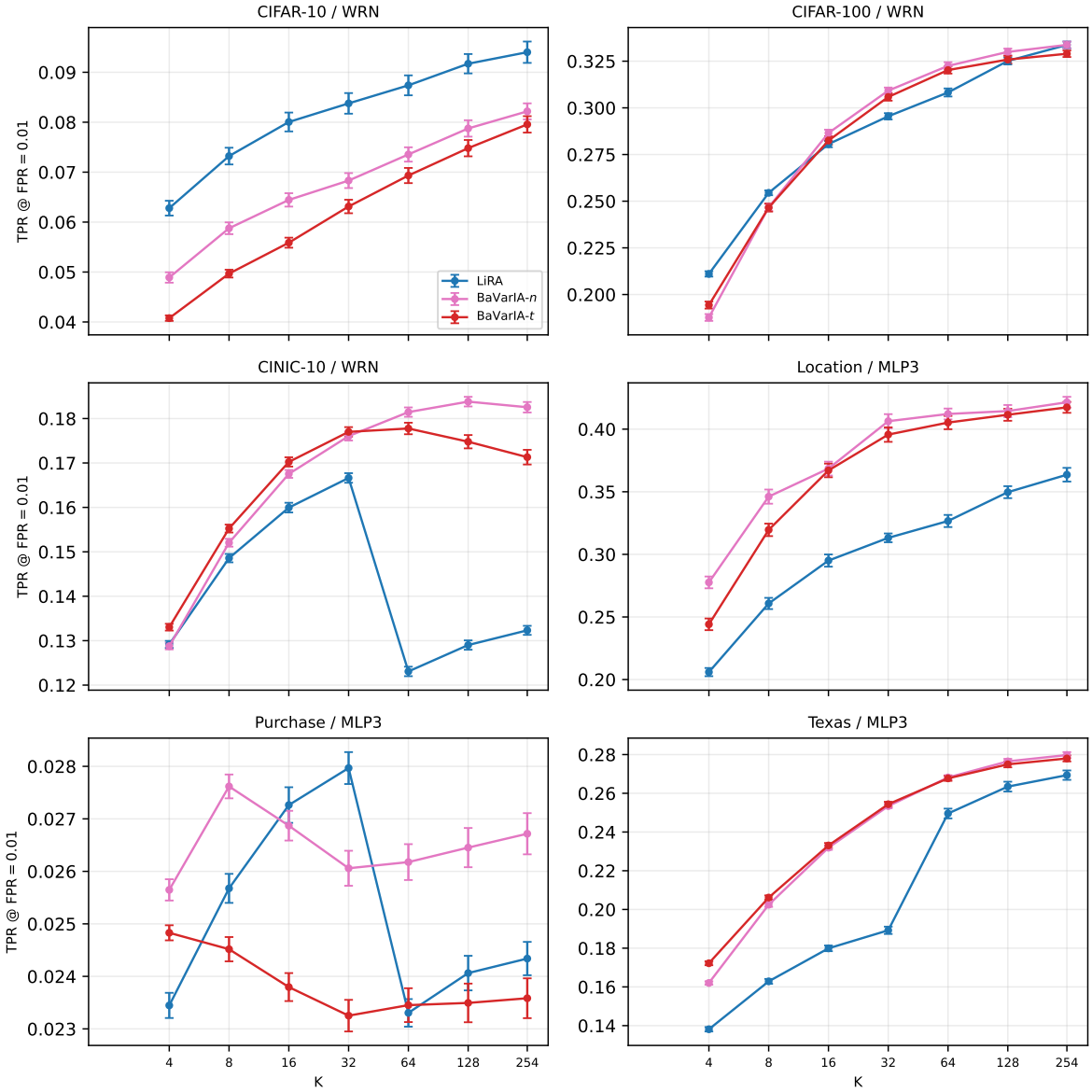


Figure 13: Offline TPR at FPR= 0.01 vs. $K$ (WRN / MLP3 testbeds). Mean ±1 SE over 32 replicates.

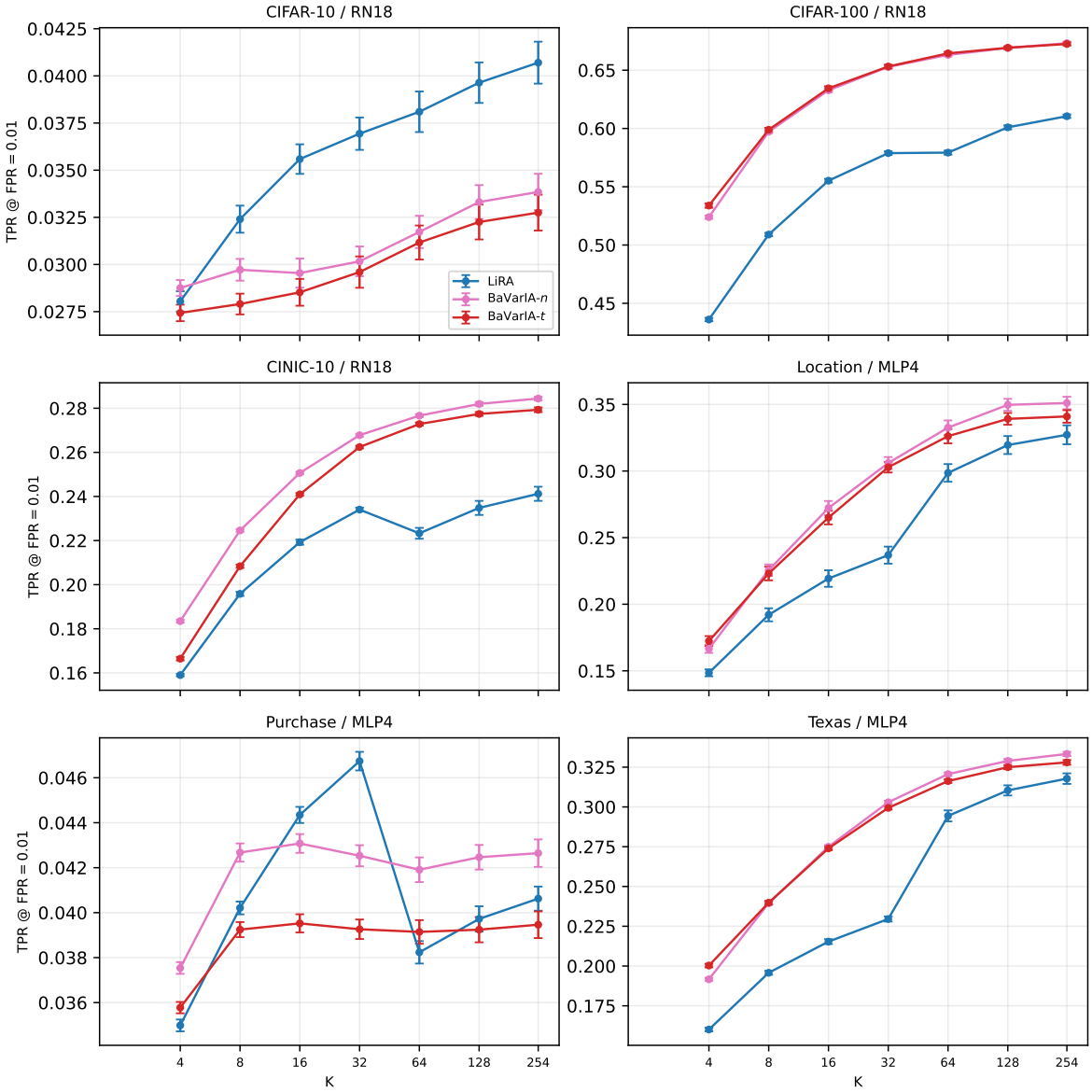


Figure 14: Offline TPR at FPR= 0.01 vs. $K$ (RN18 / MLP4 testbeds). Mean ±1 SE over 32 replicates.

**Offline results – ROC curves**

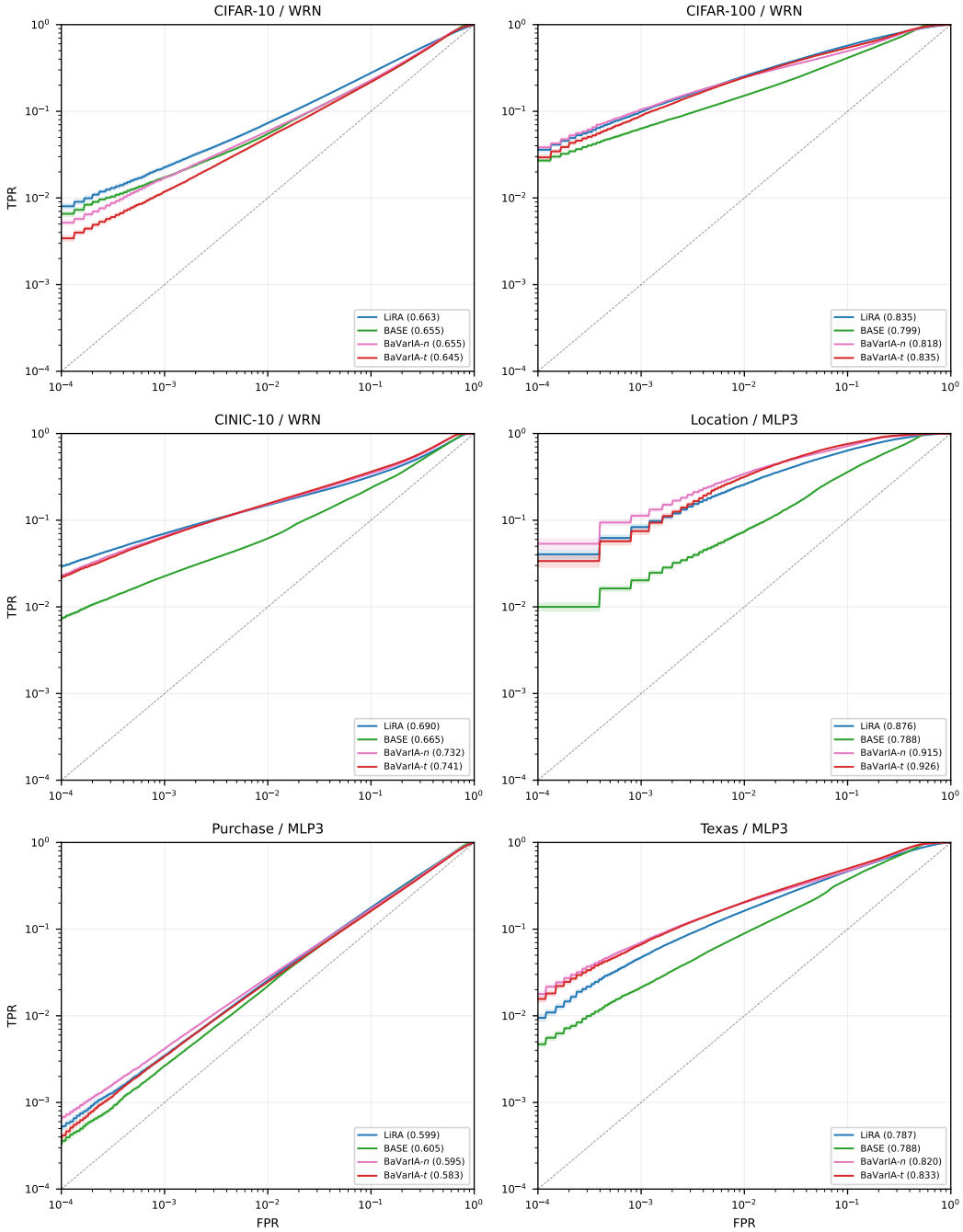


Figure 15: Offline ROC at $K = 8$ (WRN / MLP3 testbeds). LiRA, BASE, BaVarIA-$n$, BaVarIA-$t$ overlaid.

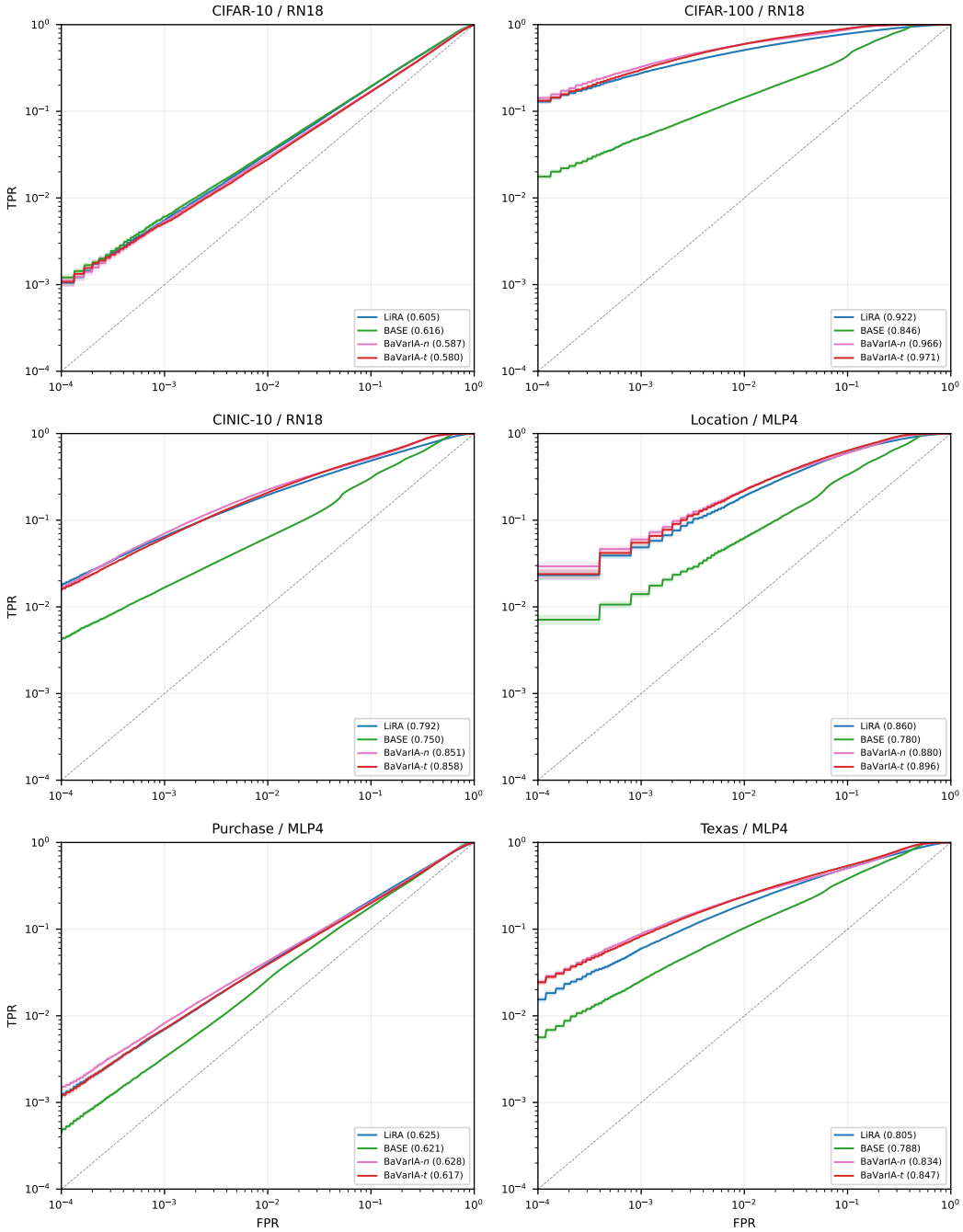


Figure 16: Offline ROC at $K = 8$ (RN18 / MLP4 testbeds). LiRA, BASE, BaVarIA-$n$, BaVarIA-$t$ overlaid.

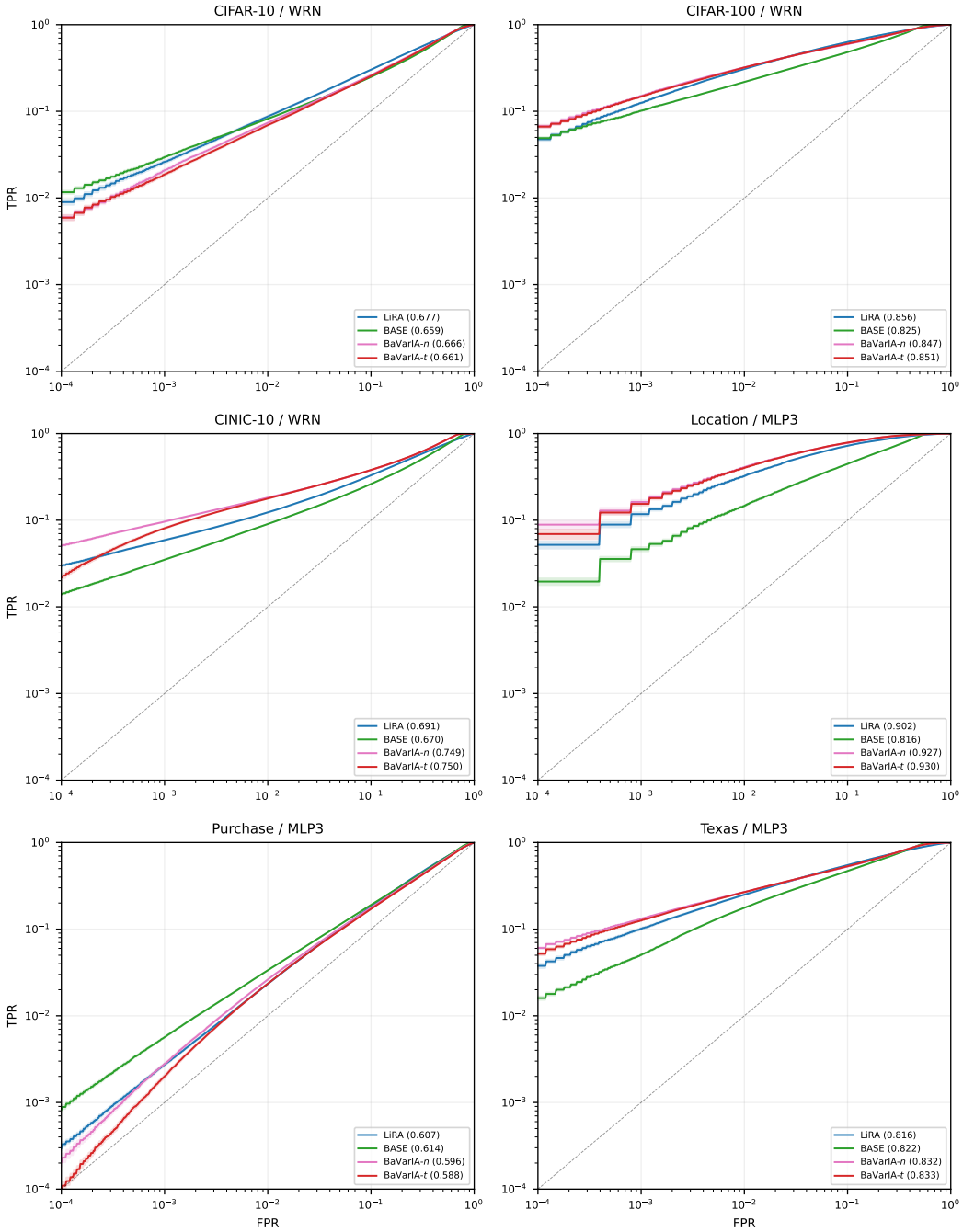


Figure 17: Offline ROC at $K = 64$ (WRN / MLP3 testbeds). LiRA, BASE, BaVarIA-$n$, BaVarIA-$t$ overlaid.

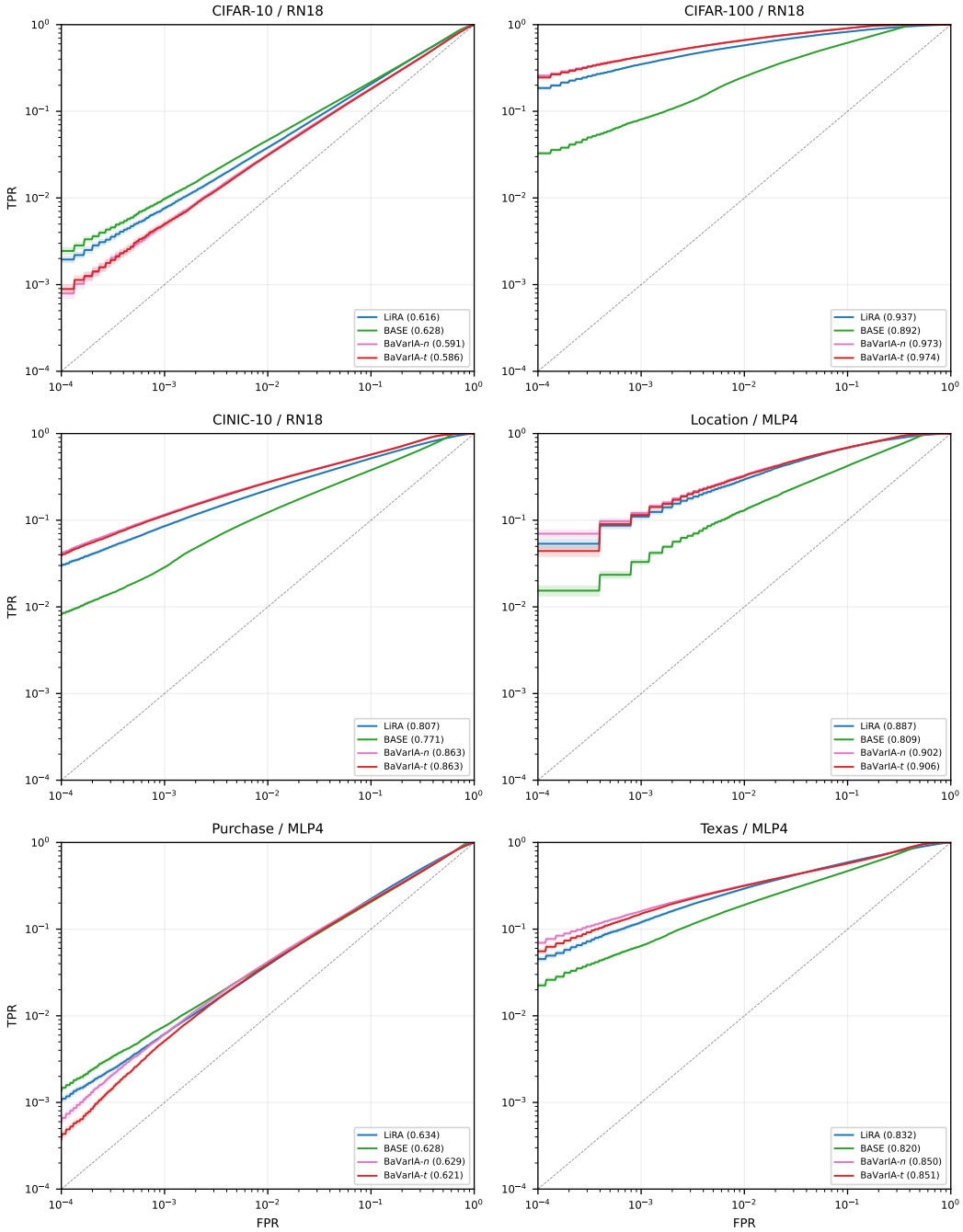


Figure 18: Offline ROC at $K = 64$ (RN18 / MLP4 testbeds). LiRA, BASE, BaVarIA-$n$, BaVarIA-$t$ overlaid.

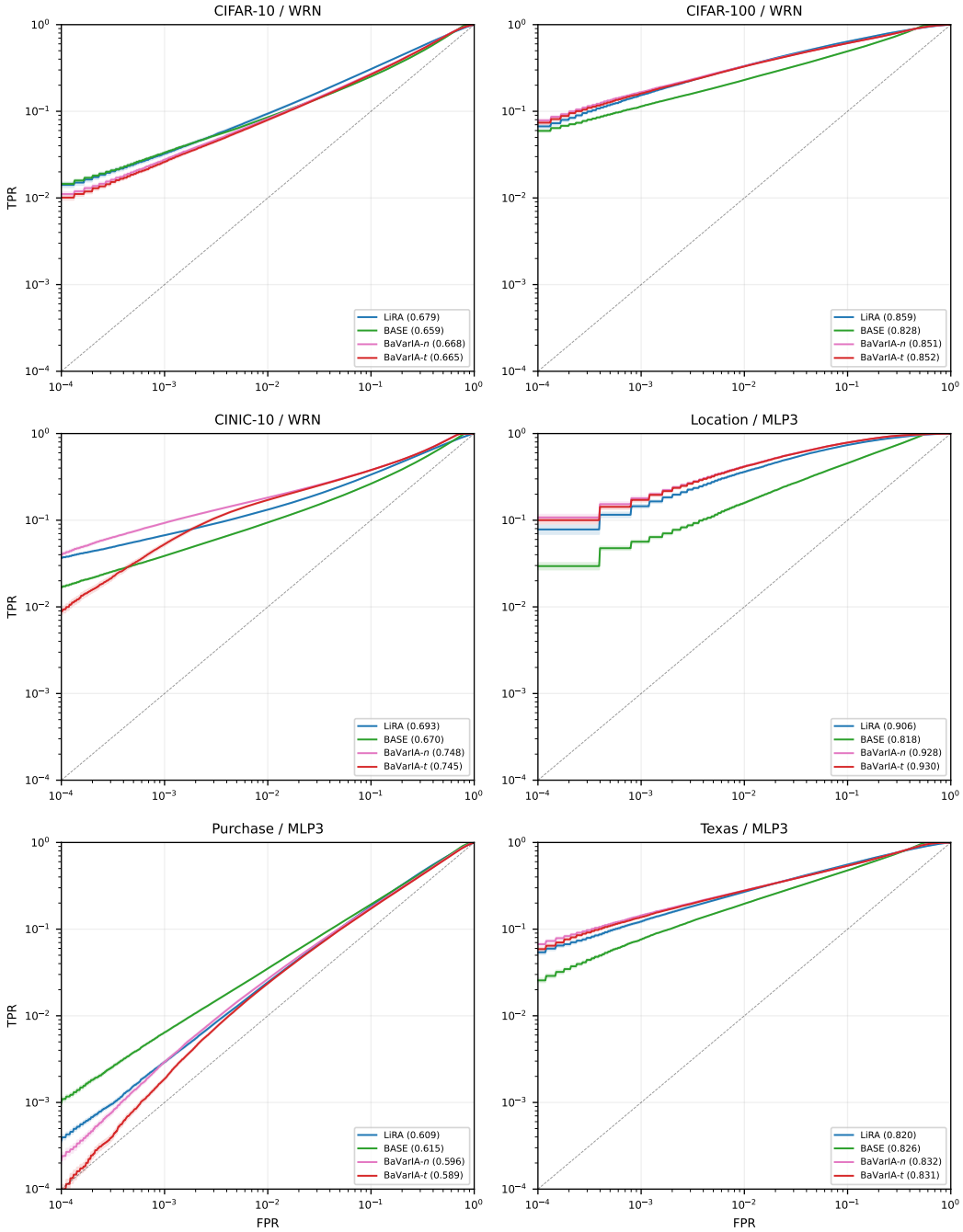


Figure 19: Offline ROC at $K = 254$ (WRN / MLP3 testbeds). LiRA, BASE, BaVarIA-$n$, BaVarIA-$t$ overlaid.

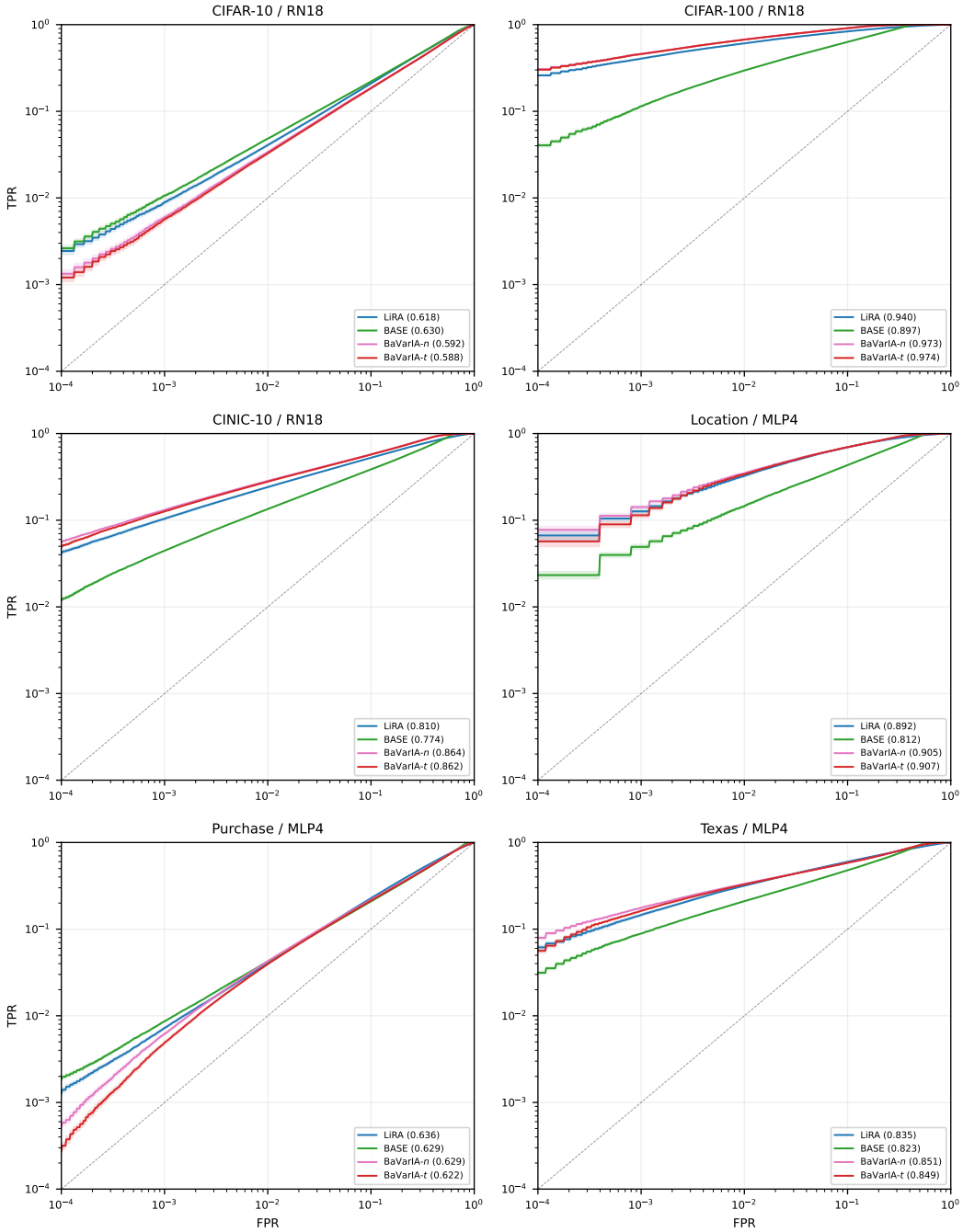


Figure 20: Offline ROC at $K = 254$ (RN18 / MLP4 testbeds). LiRA, BASE, BaVarIA-$n$, BaVarIA-$t$ overlaid.

### Offline results – per-testbed tables

| Testbed | LiRA | BaVarIA-$n$ | BaVarIA-$t$ |
|---|---|---|---|
| CIFAR-10 / WRN | **0.663** | 0.655 | 0.645 |
| CIFAR-100 / WRN | **0.835** | 0.818 | **0.835** |
| CINIC-10 / WRN | 0.690 | 0.732 | **0.741** |
| Location / MLP3 | 0.876 | 0.915 | **0.926** |
| Purchase / MLP3 | **0.599** | 0.595 | 0.583 |
| Texas / MLP3 | 0.787 | 0.820 | **0.833** |
| CIFAR-10 / RN18 | **0.605** | 0.587 | 0.580 |
| CIFAR-100 / RN18 | 0.922 | 0.966 | **0.971** |
| CINIC-10 / RN18 | 0.792 | 0.851 | **0.858** |
| Location / MLP4 | 0.860 | 0.880 | **0.896** |
| Purchase / MLP4 | 0.625 | **0.628** | 0.617 |
| Texas / MLP4 | 0.805 | 0.834 | **0.847** |

Table 7: Offline AUC, $K = 8$. Mean over 32 replicates. Best per row in bold.

| Testbed | LiRA | BaVarIA-$n$ | BaVarIA-$t$ |
|---|---|---|---|
| CIFAR-10 / WRN | **0.677** | 0.666 | 0.661 |
| CIFAR-100 / WRN | **0.856** | 0.847 | 0.851 |
| CINIC-10 / WRN | 0.691 | 0.749 | **0.750** |
| Location / MLP3 | 0.902 | 0.927 | **0.930** |
| Purchase / MLP3 | **0.607** | 0.596 | 0.588 |
| Texas / MLP3 | 0.816 | 0.832 | **0.833** |
| CIFAR-10 / RN18 | **0.616** | 0.591 | 0.586 |
| CIFAR-100 / RN18 | 0.937 | 0.973 | **0.974** |
| CINIC-10 / RN18 | 0.807 | **0.863** | 0.863 |
| Location / MLP4 | 0.887 | 0.902 | **0.906** |
| Purchase / MLP4 | **0.634** | 0.629 | 0.621 |
| Texas / MLP4 | 0.832 | 0.850 | **0.851** |

Table 8: Offline AUC, $K = 64$. Mean over 32 replicates. Best per row in bold.

| Testbed | LiRA | BaVarIA-$n$ | BaVarIA-$t$ |
|---|---|---|---|
| CIFAR-10 / WRN | **0.679** | 0.668 | 0.665 |
| CIFAR-100 / WRN | **0.859** | 0.851 | 0.852 |
| CINIC-10 / WRN | 0.693 | **0.748** | 0.745 |
| Location / MLP3 | 0.906 | 0.928 | **0.930** |
| Purchase / MLP3 | **0.609** | 0.596 | 0.589 |
| Texas / MLP3 | 0.820 | **0.832** | 0.831 |
| CIFAR-10 / RN18 | **0.618** | 0.592 | 0.588 |
| CIFAR-100 / RN18 | 0.940 | 0.973 | **0.974** |
| CINIC-10 / RN18 | 0.810 | **0.864** | 0.862 |
| Location / MLP4 | 0.892 | 0.905 | **0.907** |
| Purchase / MLP4 | **0.636** | 0.629 | 0.622 |
| Texas / MLP4 | 0.835 | **0.851** | 0.849 |

Table 9: Offline AUC, $K = 254$. Mean over 32 replicates. Best per row in bold.

| Testbed | LiRA | BaVarIA-$n$ | BaVarIA-$t$ |
|---|---|---|---|
| CIFAR-10 / WRN | **0.073** | 0.059 | 0.050 |
| CIFAR-100 / WRN | **0.254** | 0.247 | 0.247 |
| CINIC-10 / WRN | 0.149 | 0.152 | **0.155** |
| Location / MLP3 | 0.261 | **0.346** | 0.320 |
| Purchase / MLP3 | 0.026 | **0.028** | 0.025 |
| Texas / MLP3 | 0.163 | 0.202 | **0.206** |
| CIFAR-10 / RN18 | **0.032** | 0.030 | 0.028 |
| CIFAR-100 / RN18 | 0.509 | 0.597 | **0.599** |
| CINIC-10 / RN18 | 0.196 | **0.225** | 0.208 |
| Location / MLP4 | 0.192 | **0.226** | 0.223 |
| Purchase / MLP4 | 0.040 | **0.043** | 0.039 |
| Texas / MLP4 | 0.196 | 0.240 | **0.240** |

Table 10: Offline TPR @ FPR= 0.01, $K = 8$. Mean over 32 replicates. Best per row in bold.

| Testbed | LiRA | BaVarIA-$n$ | BaVarIA-$t$ |
|---|---|---|---|
| CIFAR-10 / WRN | **0.087** | 0.074 | 0.069 |
| CIFAR-100 / WRN | 0.308 | **0.322** | 0.320 |
| CINIC-10 / WRN | 0.123 | **0.181** | 0.178 |
| Location / MLP3 | 0.327 | **0.412** | 0.405 |
| Purchase / MLP3 | 0.023 | **0.026** | 0.023 |
| Texas / MLP3 | 0.250 | **0.268** | 0.268 |
| CIFAR-10 / RN18 | **0.038** | 0.032 | 0.031 |
| CIFAR-100 / RN18 | 0.579 | 0.663 | **0.664** |
| CINIC-10 / RN18 | 0.223 | **0.277** | 0.273 |
| Location / MLP4 | 0.299 | **0.333** | 0.326 |
| Purchase / MLP4 | 0.038 | **0.042** | 0.039 |
| Texas / MLP4 | 0.294 | **0.321** | 0.316 |

Table 11: Offline TPR @ FPR= 0.01, $K = 64$. Mean over 32 replicates. Best per row in bold.

| Testbed | LiRA | BaVarIA-$n$ | BaVarIA-$t$ |
|---|---|---|---|
| CIFAR-10 / WRN | **0.094** | 0.082 | 0.080 |
| CIFAR-100 / WRN | **0.334** | **0.334** | 0.329 |
| CINIC-10 / WRN | 0.132 | **0.183** | 0.171 |
| Location / MLP3 | 0.364 | **0.421** | 0.417 |
| Purchase / MLP3 | 0.024 | **0.027** | 0.024 |
| Texas / MLP3 | 0.269 | **0.280** | 0.278 |
| CIFAR-10 / RN18 | **0.041** | 0.034 | 0.033 |
| CIFAR-100 / RN18 | 0.611 | **0.673** | 0.672 |
| CINIC-10 / RN18 | 0.241 | **0.284** | 0.279 |
| Location / MLP4 | 0.327 | **0.351** | 0.341 |
| Purchase / MLP4 | 0.041 | **0.043** | 0.039 |
| Texas / MLP4 | 0.318 | **0.333** | 0.328 |

Table 12: Offline TPR @ FPR= 0.01, $K = 254$. Mean over 32 replicates. Best per row in bold.